

The Kelly Criterion: Optimal Stochastic Control in Fixed-Limit Games

Team Odessa

December 23, 2025

1. Theoretical Context: Why Kelly?

In imperfect information games like poker, the objective is often misidentified as maximizing *Expected Value* (EV). However, EV maximization assumes infinite bankroll and time. In a finite tournament, a strategy that maximizes EV but carries high variance can lead to **Ruin** (an insurmountable point deficit) before the long run is realized.

The **Kelly Criterion** solves this by maximizing the **Geometric Growth Rate** of the bankroll (score). It mathematically guarantees:

1. **Maximum Wealth Velocity:** It reaches a target score faster than any other strategy.
2. **Ruin Prevention:** It implicitly assigns a value of $-\infty$ to bankruptcy, ensuring the agent never risks more than is mathematically justifiable by its edge.

2. Mathematical Derivation

Let W_0 be the initial wealth. We wager a fraction f on a repeated bet with probability of winning p , probability of losing $q = 1 - p$, and payoff odds b (we win b units for every 1 unit risked). After N trials with W wins and L losses:

$$W_N = W_0(1 + fb)^W(1 - f)^L$$

To find the optimal f , we maximize the **Expected Exponential Growth Rate** $G(f)$:

$$G(f) = \mathbb{E} \left[\ln \left(\frac{W_N}{W_0} \right)^{\frac{1}{N}} \right] = p \ln(1 + fb) + q \ln(1 - f)$$

We find the critical point by setting the derivative with respect to f to zero:

$$\frac{dG}{df} = \frac{pb}{1 + fb} - \frac{q}{1 - f} = 0$$

$$pb(1 - f) = q(1 + fb) \implies pb - pbf = q + qfb$$

Substituting $q = 1 - p$:

$$pb - (1 - p) = f(pb + b(1 - p)) \implies p(b + 1) - 1 = fb$$

$$f^* = \frac{p(b + 1) - 1}{b}$$

3. Application to Finopoly (Discrete Betting Context)

In Finopoly, we face a **Fixed-Bet Constraint**. We cannot adjust our bet size to match f^* . We must risk either 0 points (Fold) or 3 points (Raise). We resolve this by converting the *Kelly Size* into a *Kelly Frequency*.

3.1 Establishing the Odds (b)

We define risk relative to the **Baseline of Folding** (Sunk Cost of -1).

- **Upside:** Raising wins +3. Relative Gain: $(+3) - (-1) = +4$.
- **Downside:** Raising loses -3. Relative Risk: $(-1) - (-3) = +2$.
- **Payoff Odds (b):** $\frac{\text{Gain}}{\text{Risk}} = \frac{4}{2} = 2$.

3.2 The Frequency Transformation

Substituting $b = 2$ into the Kelly equation yields the optimal risk exposure:

$$f^* = \frac{p(3) - 1}{2}$$

Since the game enforces a discrete bet size B_{fixed} , we define our Raise Probability $P(R)$ such that our **Expected Risk Exposure** equals the Kelly Optimum:

$$\mathbb{E}[\text{Risk}] = P(R) \cdot B_{fixed} \equiv f^* \cdot \text{Bankroll}$$

Assuming a standardized risk unit ($B_{fixed} \approx \text{Bankroll}$ for the local decision), we derive the **Odessa Stochastic Gate**:

$$P(\text{Raise}) = \max\left(0, \frac{3p - 1}{2}\right)$$

This transformation ensures that while we cannot optimize *size*, our long-term variance exposure matches the optimal growth curve of the Kelly criterion.

4. Computational Implementation

The following Python snippet implements the derived Odessa Stochastic Gate. It calculates f^* and uses a random number generator to execute the mixed strategy frequency.

```

1 import random
2
3 def get_kelly_decision(win_prob):
4     b_odds = 2.0
5     f_star = (win_prob * (b_odds + 1) - 1) / b_odds
6
7     if f_star <= 0:
8         return False
9
10    return random.random() < f_star

```

Listing 1: Odessa Kelly Implementation